2.4 Step 2: Landslide susceptibility assessment

2.4.1 Objective

The goal of this step is to assess where landslides may occur in the study areas and to produce, for the four sub-catchments, landside susceptibility maps.

2.4.2 Method

2.4.2.1 The role of the landslide inventories

The number of inventoried landslides in Step 1 is much too limited for applying any data-driven susceptibility and hazard assessment at the scale of the four, relatively small, sub-catchments (Corominas et al., 2014; Reichenbach et al., 2018; Depicker et al., 2020b). Therefore, to specifically compute data-driven models and meet the commitments that we could have expected if more landslides were present in the sub-catchments, we decided to use, as an extra, an existing regional inventory of shallow landslides that covers the Lake Kivu region. In Rwanda, this inventory covers all the regions west of Kigali (Figure 3). The inventory has already been used for several susceptibility and hazard assessments in the region (Depicker et al. 2020a, 2020b) and should be made publicly available once a manuscript currently in review (Depicker et al., in review) is published. From this inventory, it is possible to calibrate regional data-driven susceptibility (and subsequently hazard) models that cover the four sub-catchments at once. Since the assessments are made at the regional level, it also allows us to compare the four study sites with each other.

In this study, we therefore used two inventories for two types of susceptibility assessments, one produced specifically here (Inventory 1 – see Step 1 for details) and one from other works (Inventory 2):

- **Inventory 1 – used for sub-catchment susceptibility to ground deformations within the old and recent deep-seated landslides:**

For the old and recent deep-seated landslides, the hazard assessment, even at the regional level, cannot be assessed without any accurate information on their timing of occurrence. For the sole susceptibility assessment, although a regional model for these landslides is already available (Depicker et al., 2020b), its use at the local scale is meaningless; a regional model for deep-seated landslides provides only an information on their general distribution patterns. At the local scale, the exact location of a future deep-seated slope failure is here impossible to predict from a data-driven model since it depends on heterogenous local conditions associated with the lithology and hydrogeology that are not known (Dille et al., 2019).

However, deep-seated landslides are places where the hillslope material has been disturbed. Therefore, the landsides are portions of hillslopes that are more easily prone to ground deformations (e.g. Nobile et al., 2018; Dille et al., 2021). For the recent landslides, such deformations are often visible and the delineation of the active zones is straightforward (Figure 2 d).

For the old landslides, although such deformations can be unnoticed to the naked eye, their effect can be pervasive and, on a relatively long period, impacts the infrastructures (fissures, cracks).